Yifan Feng

Ph.D. Candidate in Entomology — mosquito genomics, chromosomal inversions, and cytogenetics

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Education

PhD **Virginia Polytechnic Institute and State University**, Entomology — Blacksburg, VA — Jan 2022 – Dec 2026

- Advisor: Dr. Maria V. Sharakhova
- Research: Chromosomal inversions and genome evolution in *Culex* and *Aedes* mosquitoes
- Department of Entomology and Fralin Life Sciences Institute
- Expected completion: December 2026; dissertation defense scheduling in progress
- Coursework: Molecular Genetics, Bioinformatics, Genomics, Biochemistry, Experimental Design

BS **Virginia Polytechnic Institute and State University**, Biological Sciences — Blacksburg, VA — Aug 2017 – May 2021

BS **Virginia Polytechnic Institute and State University**, Food Science and Technology — Blacksburg, VA — Aug 2017 – May 2021

Research Experience

Virginia Tech — Department of Entomology & Fralin Life Sciences Institute, Graduate Research Assistant — Blacksburg, VA — Jan 2022 – present

- Prepared **Hi-C libraries** for mosquito genome and structural variation studies; coordinated sequencing requirements, sample metadata, and data delivery with sequencing providers.
- Contributed to a **haplotype-resolved *Aedes albopictus* genome assembly** integrating **trio-binning**, **Nanopore long reads**, **Hi-C scaffolding**, and **FISH** physical validation; first author of the manuscript in preparation.
- Designed and validated **PCR breakpoint assays** to genotype chromosomal inversions across natural populations, linking structural variants to climatic adaptation and vector traits.
- Conducted population genomic analyses (**FST, PCA, AMOVA**) to quantify differentiation and gene flow among *Culex pipiens* populations across North America.
- Performed molecular dissections, DNA extraction, and fluorescence microscopy imaging; maintained mosquito colonies and coordinated laboratory safety inspection responses.
- Mentored two undergraduate researchers in mosquito dissection, FISH probe preparation, and PCR optimization and troubleshooting for mosquito identification and population studies.
- Contributed to the international *Aedes aegypti* inversion study published in *Genome Biology and Evolution*.
- Prepared research summaries and FISH figures for an NIH grant progress report, incorporating feedback from the principal investigator.

Virginia Tech — Sharakhova Lab, Research Assistant (before doctoral enrollment) — Blacksburg, VA — Oct 2021 – Dec 2021

- Supported mosquito colony maintenance and molecular cytogenetics, including PCR, DNA probe labeling, chromosome preparation, and FISH.
- Joined the laboratory in fall 2021 before beginning the PhD in spring 2022.

Virginia Tech — Department of Food Science and Technology, Undergraduate Researcher — Blacksburg, VA
Aerosolized *Listeria* contamination from cavitation treatment of raw produce. — Sept 2020 – Dec 2020

- Applied microbubble cavitation treatments and quantified resulting aerosolized bacterial contamination.
- Built MATLAB models of cavitation effects and ran predictive simulations on university high-performance computing resources.
- Compiled, visualized, and statistically analyzed experimental results to identify contamination trends.

Publications

Chromosomal Inversions and Their Potential Impact on the Evolution of Arboviral Vector *Aedes aegypti* — July 2025
Jiangtao Liang, Noah H. Rose, Ilya I. Brusentsov, Varvara Lukyanchikova, Dmitriy A. Karagodin, *Yifan Feng*, et al.
10.1093/gbe/evaf118 (Genome Biology and Evolution 17(7), evaf118)

Hi-C Proximity Ligation Approach Identified Chromosomal Rearrangements in *Culex pipiens* Mosquitoes — 2025
Yifan Feng, Varvara Lukyanchikova, Jiangtao Liang, Dmitriy A. Karagodin, Ilya I. Brusentsov, Megan L. Fritz, et al.
(American Journal of Tropical Medicine and Hygiene 112(6) — ASTMH Annual Meeting abstract)

Chromosomal Rearrangements Differentiate Species of the *Culex pipiens* Complex — 2025
Maria V. Sharakhova, *Yifan Feng*, Varvara Lukyanchikova, Jiangtao Liang, Dmitriy A. Karagodin, et al.
(Conference proceedings)

Chromosomal Inversions Differentiate Mosquitoes in the *Culex pipiens* Complex — 2026
Yifan Feng, Varvara Lukyanchikova, Jiangtao Liang, Ilya I. Brusentsov, Dmitriy A. Karagodin, Megan L. Fritz, Maria V. Sharakhova
(Manuscript in preparation)

A haplotype-resolved genome assembly enables the discovery of chromosomal inversions in the arboviral vector *Aedes albopictus* — 2026
Yifan Feng, Chujia Chen, Varvara Lukyanchikova, Christen Hughes, Mark Potters, Ankita Sindhania, et al.
(Manuscript in preparation for bioRxiv submission)

Conference Presentations

- **Talk (accepted)** — *Hi-C proximity ligation reveals chromosomal inversions potentially associated with host preference in Culex pipiens mosquitoes.* Student 10-Minute Presentation Competition, Entomology 2026 (ESA Annual Meeting), Columbus, OH, November 2026.
- **Poster** — *Hi-C proximity ligation approach identified chromosomal rearrangements in Culex pipiens mosquitoes.* American Society of Tropical Medicine and Hygiene (ASTMH) Annual Meeting, New Orleans, LA, November 16, 2024. Poster 7815, Session C.
- **Oral presentation** — *Chromosomal inversions differentiate mosquitoes in the Culex pipiens complex.* CeZAP Infectious Diseases Symposium, Blacksburg, VA, October 6, 2023.

Awards

- **Fifth place (runner-up), 2025–2026 Student Poster Competition**, Virginia Mosquito Control Association. Awarded December 2025 for an *Aedes albopictus* genome assembly poster; complimentary registration to the 2026 VMCA conference.

Teaching and Mentoring

Virginia Tech — Department of Entomology, Graduate Teaching Assistant, ENT 2004 (Insects and Human Society) — Blacksburg, VA — Jan 2024 – Dec 2024

- Supported the asynchronous online course in Spring 2024 and the in-person course in Fall 2024, working with Dr. James Wilson and Dr. Ksenia Onufrieva.
- Graded assignments and provided rubric-based feedback in Canvas.
- Prepared short Insects in the News presentations and associated quiz or exam questions.

Sharakhova Lab, Virginia Tech, Research Mentor, Undergraduate Researchers — Blacksburg, VA — Apr 2024 – Dec 2025

- Trained two undergraduate researchers in mosquito dissection, FISH probe preparation, and PCR assays for mosquito identification and population studies.
- Troubleshot COI and ACE assays by reviewing primer dilutions, polymerase selection, reaction composition, cycling conditions, and gel results.

Technical Skills

Molecular Biology: DNA/RNA extraction, PCR, qPCR, RT-qPCR, ddPCR, gel electrophoresis, cloning, nucleic acid purification

Library Prep & Sequencing: Hi-C library preparation, Sanger sequencing, Illumina WGS sample preparation and submission

Cytogenetics & Imaging: Chromosome preparation, FISH probe synthesis and hybridization, fluorescence microscopy (Zeiss ZEN), ImageJ

Genomics & Bioinformatics: Hi-C scaffolding (Juicer, Juicebox), genome assembly from Nanopore long-read data, trio-binning, gene annotation, alignment (BWA, Minimap2)

Population Genetics: FST, PCA, AMOVA, STRUCTURE, PLINK, inversion genotyping

Cell & Assay Work: Mammalian and insect cell culture, sterile technique, ELISA, Western blot, BSL-2 practice

Programming & Software: R (ggplot2), Python, Linux/Unix, MATLAB, Galaxy, Geneious, BLAST/NCBI, GraphPad Prism, BioRender

Scientific Communication: Manuscript preparation, grant proposal writing, poster and oral presentation, mentoring, laboratory documentation

Service and Professional Membership

- Volunteer, Mosquito Genetics exhibit, Hokie BugFest 2025, Virginia Tech — public outreach through hands-on activities and chromosome models.
- Member, Entomological Society of America.
- Member, Virginia Mosquito Control Association.